Nkulukeko.
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Exerzisez

BrightLight Data Analytics

- Question 1

① Total Number of employees

- Select Count (ID) AS total number of employees
FROM. Employee_db;

outcome 10 Employees.

Employee_db.

ID	first_Name	Last Name	Department
1	John	Doe	IT
2	Jane	Smith	HR
3	Mike	Johnson	finance
4	Sarah	Brown	IT
5	David	White	Marketing
6	Emily	Davis	IT
7	Robert	Wilson	finance
8	Jessica	Moore	HR
9	Daniel	Clark	Marketing
10	Laura	Hall	IT

② Total Salary IT Department

Select Sum (Department) AS total salary IT
FROM 'employee_db';
~~Group by (Department =~~

Total 220 000

Outcome employee-db

ID	first_Name	Last name	Department	Salary
1	John	Doe	IT	55000
4	Sarah	Brown	IT	62000
6	Emily	Davis	IT	62000
10	Laura	Hall	IT	50000
				220 000

Total

③ AVG Total salary HR Department

Select SUM (Department) As Total Salary HR
FROM Employee_db;

Employee_db

ID	First Name	Last Name	Department	Salary	
2	Jane	Smith	HR	48000	
8	Jessica	Moore	HR	51000	
				99000	Total

Total Salary % Number of employees
99000 % 2
49500 Average

④ Min Max highest and lowest Salary

Min
Select min (salary) As Lowest
from 'Employee_db';

Max
Select max (salary) As highest
From 'Employee_db';

Out Come Employee_db.

ID	Firstname	Last Name	Departm	Salary	
2	Jane	Smith	HR	48000	Lowest Min
6	Emily	Davis	IT	62000	highest Max

5 Group BY statement Employees By Department
total Salary paid each Department

```
Select*  Department,
         Salary,
FROM 'Employee_db'
GroupBy  Department;
```

outcome.

ID	First Name	Last Name	Department	Salary	Totals per department
1	John	Doe	IT	55000	
4	Sarah	Brown	IT	53000	
6	Emily	Davis	IT	62000	IT
10	Laura	Hall	IT	50000	220 000
2	Jane	Smith	HR	4800	HR
8	Jessica	Moore	HR	51000	99000
3	Mike	Johnson	finance	60000	Finance
7	Robert	Wilson	finance	59000	119000
5	David	White	Marketing	52000	Marketing
9	Daniel	Clark	marketing	53000	105 000

6 Group BY and Count

```
Select Count  ID,
              City
FROM 'Employees_db'
Groupby City;
```

Employee_db

ID	~~Number~~ Number of Employees	City
1 4	2	New York
2 6 9	3	Chicago
3 8	2	Los Angeles
5 10	2	San fransicco
7	1	Houston

⑦ Group By and Order By

```
Select Department,
    SUM~~AVG~~ (Total Salary ÷ Number of Employees) As AVG Salary
FROM 'Employee_db'
Group By Salary average
ORDER BY ASEC ;
```

Average = total Salary / Number of employees.

ID	Department	Salary	Average Salary	Ascending
2 6 9 10	Chicago IT	~~45~~000 53000 62000 50000	55000 AVG	59500
2 8	HR	48000 51000	49500	55000
3 7	Finance	60000. 59000	59,500	52500
5 9	Marketing	52000 53000	~~105000~~ 52500	49500

8 Having Clause

```
Select Department,
   SUM (Salary) AS Total Salary
FROM 'Employee_db'
Group By Department;
```

outcome.

Department	total Salary exceeding 100000
IT	220 000
Finance	119000
Marketing	105 600

9 Combining Group By, Having and ORDER BY Number of employees decending

```
Select sum(City) As more than One Employees
FROM 'employee_db'
ORDER BY DESC;
```

Outcome.

City	Number of Employees	Descending
Chicago	3	3
NEW YORK	2	2
Los Angeles	2	2
San fransisco	2	2
New York	2	2

10 Department with highest average Salary.

```
Select SUM $alary AS highest Average Salary
FROM 'Employee_db'
GROUP By Department;
```

ID	Department	Salary	Average
3	Finance	60000	59500
7		59000	
	Total	119000	

AVG = total salary per Department / Number of employees.
= 119000 / 2 employees
59500